



Midterm Assignment: Prompt Engineering

WEEKS 1–5 COMPREHENSIVE ASSESSMENT

PROMPT ENGINEERING COURSE

ANKARA MEDIPOL UNIVERSITY

AYŞE EBRAR KALKIN ID:11230070 SUBMISSION DATE: 15.11.2025

Part 1: Prompt Comparison Analysis

a. Domain & Task Description (15 words)

Chosen Domain: Healthcare

Specific Task: Draft an informational handout for family caregivers about Alzheimer's disease.

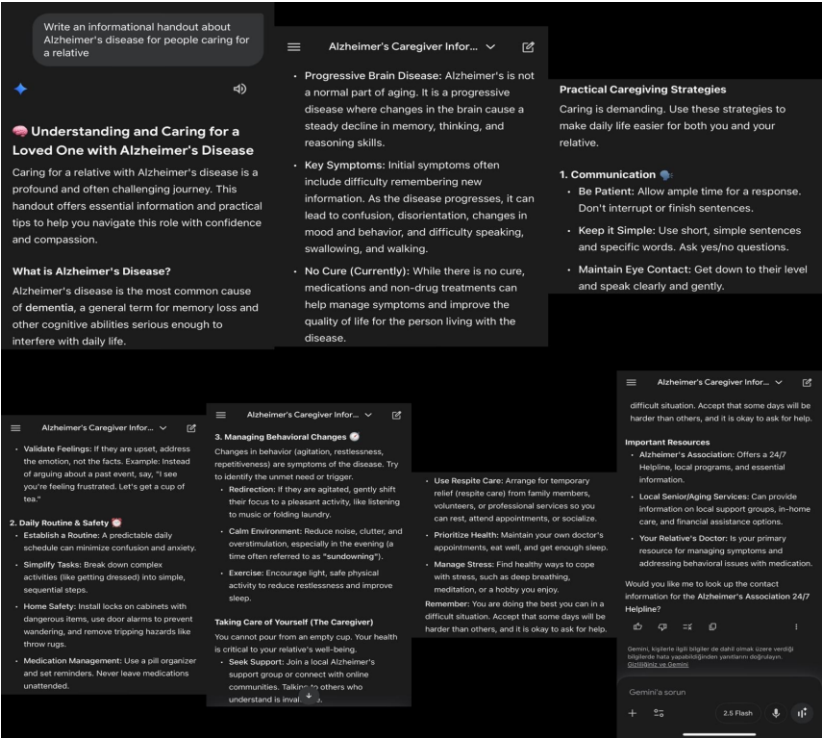
b. Version A: Vague Prompt

Vague Prompt: "Write an informational handout about Alzheimer's disease for people caring for a relative"

AI Tool Used: Gemini 2.5 Flash

AI Response:

Combined Screenshots: AI response to my vague prompt.



c. Version B: Improved Prompt

Improved Prompt: "You are an Empathy-driven Dementia Support Specialist (Role Prompting). Your task is to create a guide on Alzheimer's disease for first-time family caregivers (Audience). The content must be supportive, non-technical, and highly practical (Tone/Constraints). The output must be formatted as a two-column markdown table (Structured Output). The table should have the following exact headings: 'Challenge' and 'Caregiver Strategy'. Include a mandatory disclaimer at the end: 'This guide is for emotional support and general information only.' (Guardrail)."

Techniques Used: Role Prompting, Audience Specification (first-time family caregivers), Tone Specification (supportive, practical), Structured Output (two-column table, specific headings), Guardrail (mandatory disclaimer)

AI Tool Used: Gemini 2.5 Flash

AI Response:

Combined Screenshots: AI response to my improved prompt.

You are an Empathy-driven Dementia Support Specialist. Your task is to create a guide on Alzheimer's disease for first-time family caregivers. The content must be supportive, non-technical, and highly practical. The output must be formatted as a two-column markdown table. The table should have the following exact headings: 'Challenge' and 'Caregiver Strategy'. Include a mandatory disclaimer at the end. This guide is for emotional support and general information only.

📖 A Caregiver's Compass: Navigating Alzheimer's Disease

Welcome to your new role as a caregiver. It is one of the hardest, yet most meaningful, jobs you will ever have. This guide focuses on common challenges and provides practical, heart-centered strategies to help you and your loved one.

Challenge	Caregiver Strategy
Memory Loss & Repetition (Being asked the same question minutes apart, forgetting recent events.)	Respond with Kind Logic: Focus on the emotion behind the question. Ask about a loved one passed, say, "You miss them very much." Keep answers brief and gentle. Use distraction or redirection to change the topic after the first response.
Communication Breakdown (Difficulty finding words, struggling to understand complex sentences, or becoming easily frustrated when speaking.)	Simplify and Slow Down: Use short, concrete sentences and ask yes/no questions. Speak slowly and maintain calm eye contact. If they are struggling to find a word, gently suggest a word, but don't pressure them.
Wandering or Pacing (Trying to leave the house, seeming restless, especially in the late afternoon/evening—often called "sundowning.")	Address Unmet Needs: Wandering often means they are looking for something (comfort, an activity, or a need like the bathroom). Ensure they have eaten, are comfortable, and use safe physical activity (like a short walk) to ease restlessness. Install alarms on doors.
Agitation or Anger (Getting easily upset, frustrated, or aggressive.)	Look for the Trigger: Agitation is almost always caused by something: pain, being too cold/hot, overstimulation (too much noise or company), or an expectation they cannot meet. Remove the trigger, speak in a soft, low voice, and use redirection to a soothing activity (music, a favorite snack).
Refusal to Bathe or Change Clothes (Resisting necessary personal care tasks.)	Try Timing and Technique: Respect their dignity. Find the "best time of day" when they are calmest. Use no threatening language of "bath," say "a nice soak to relax." Use baths or dry shampoo if shower causes too much stress.

Alzheimer's Caregiver Information

Your Own Isolation and Exhaustion (Feeling overwhelmed, lonely, and physically drained.)

Schedule Respite and Support: Your well-being is not a luxury, it's essential. Accept help when offered. Join a local caregiver support group—connecting with others who get it is vital. Schedule at least one hour a week to do something just for you.

⚠️ Mandatory Disclaimer: This guide is for emotional support and general information only. It is not a substitute for professional medical advice. Please consult your loved one's physician or a dementia specialist for diagnosis, treatment, and specific care planning.

d. Comparison Analysis (243 words)

The objective was to compare the quality, usefulness, and specificity of the outputs generated by the vague prompt (Version A) and the improved prompt (Version B) on the topic of Alzheimer's disease for family caregivers.

Version A's output produced a generic, multi-section text block. While it covered medical facts and stages, it lacked immediate utility for a caregiver seeking quick support. The output used moderate-to-high linguistic register and presented its advice in dense paragraph form, failing to deliver the necessary scannability for a stressed audience.

On the other hand, Version B's output demonstrated superior quality, usefulness and specificity. The prompt successfully utilized Role Prompting and Structured Output.

Three concrete improvements in Version B's output are:

1.Actionability and Structure: The output strictly adhered to the structural contract by presenting information in a two-column table. This format delivers immediate, practical insights by directly pairing a problem with a solution.

2.Audience-Appropriate Tone: The language is consistently supportive and non-technical, successfully aligning the content with the specified caregiver audience and tone.

3.Safety Compliance: The output reliably included the mandated guardrail/safety disclaimer, a critical element for sensitive medical information.

The improved prompt worked better because it adhered to Prompting Principles. By being explicit about the desired Role and Constraining the Output to a specific table format, the prompt successfully focused the LLM's vast knowledge, acting like a lens to produce a precise response rather than a generic summary.

Part 2: Practical Prompt Design

a. Framework Selection & Justification (98 words)

Framework Chosen: CO-STAR. (Context, Objective, Style, Tone, Audience, Response).

Business Task: Design a customer onboarding email sequence (3 emails).

The CO-STAR Framework is the most appropriate choice for this task because generating marketing or customer-facing communication relies on defining the soft constraints of Style, Tone, and Audience.

An email sequence requires consistent Tone and a defined Style across multiple steps to achieve its ultimate Objective. By demanding explicit instructions for these parameters, CO-STAR helps avoid the anti-pattern of a vague audience and guarantees a high-quality output that aligns with professional communication goals.

b. Complete Prompt Following Framework Structure

Prompt for Initial AI Output:

[CONTEXT]: The client is an e-learning platform ("SOMC") selling short online medical courses. The user signed up for a 7-day free trial but has not yet completed their first lesson. The goal is a sequence of **3 emails** to be sent during the trial period.

[OBJECTIVE]: Design the complete 3-email sequence. The purpose is to motivate users to engage with the trial and increase conversion to a paid subscription after 7 days. Each email must be brief and have a single, clear **Call to Action (CTA)**.

[STYLE]: Conversational and personal (use second-person perspective: "you"). Use short, action-oriented sentences (average length 12-18 words). **Strictly avoid** passive voice and corporate jargon.

[TONE]: Friendly, supportive, and encouraging. Use of contractions is acceptable for a casual register. Urgency cues should be limited to Email 3.

[AUDIENCE]: Busy professionals (adults aged 25-45) looking to quickly acquire new skills. The content should use simple language and prioritize user benefit.

[RESPONSE]: Present the full sequence output as a Markdown **table** with the following exact column headings:

Email # (1, 2, 3)

Subject Line (Max 7 words)

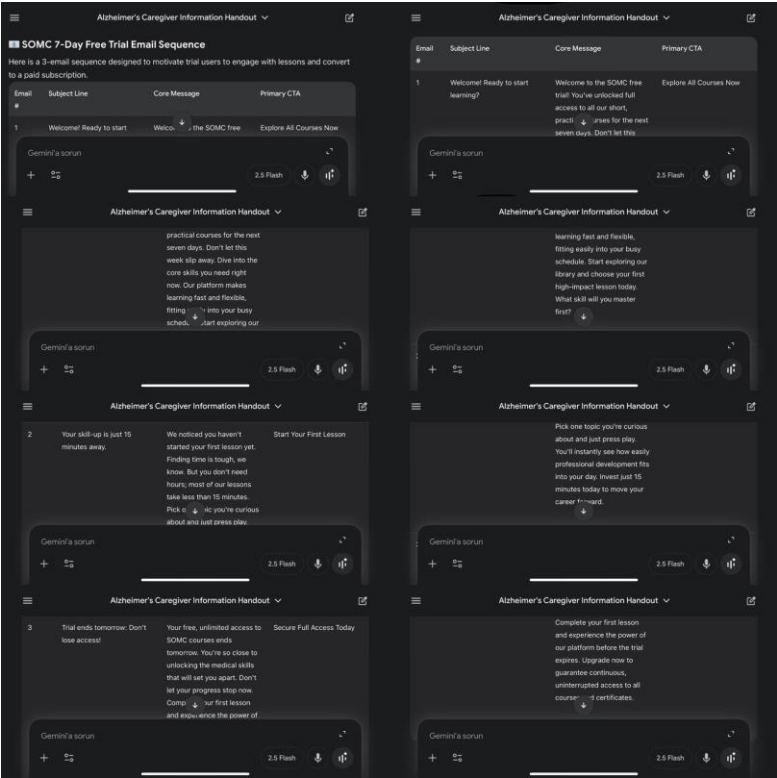
Core Message (Max 100 words)

Primary CTA (Call to Action)

c. AI Output

1. AI Tool Used: : Gemini 2.5 Flash

Combined Screenshots: AI response to my complete prompt following framework structure.



2. Highlighting AI Adherence to the CO-STAR Framework

The AI successfully adhered to the constraints specified in the [RESPONSE], [TONE], and [STYLE] components, ensuring the output was structured, brief, and supportive.

d. Refinement & Iteration (253 words)

Initial Weaknesses:

1. Subject Line Effectiveness: The Subject Lines in the initial output are somewhat generic. They fail to maximize engagement by lacking specific benefit-driven language or a strong curiosity trigger.

2. Conversion Urgency: While Email 3 introduces urgency, the final CTA could be strengthened by exploiting loss aversion and making the limited-time discount more explicit in the subject line.

Refined follow-up prompt:

"Critique the previous table output and revise it using the ****Critique-then-revise**** pattern. Focus your revisions on improving the persuasive power of the content.

Specifically:

1. Revise all ****Subject Lines**** to be either more ****curiosity-driven**** (e.g., using a short question) or ****benefit-driven****. For example, Email 2 could reference a key time constraint for busy professionals ("15 minutes").

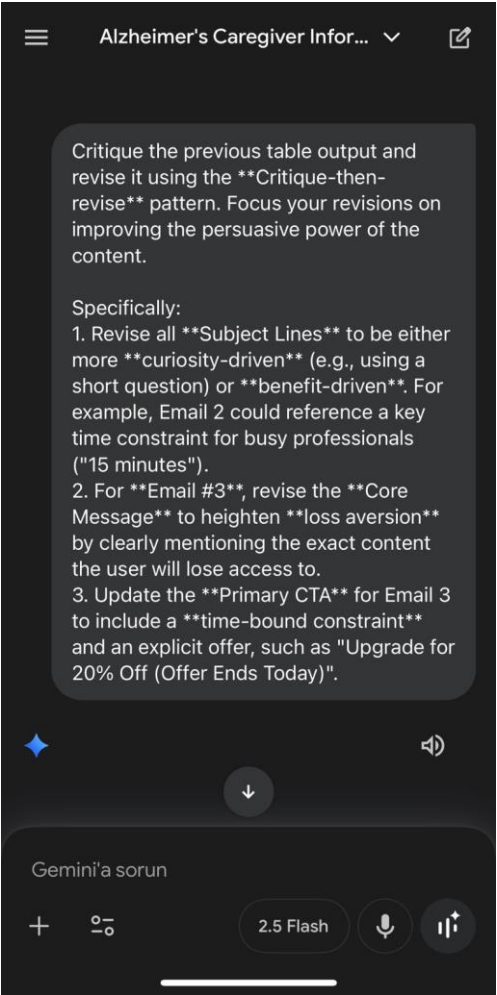
2. For ****Email #3****, revise the ****Core Message**** to heighten ****loss aversion**** by clearly mentioning the exact content the user will lose access to.

3. Update the ****Primary CTA**** for Email 3 to include a ****time-bound constraint**** and an explicit offer, such as "Upgrade for 20% Off (Offer Ends Today)".

AI response:

AI Tool Used : Gemini 2.5 Flash

Combined Screenshots: AI response to my complete prompt following framework structure after refinement & iteration.



Alzheimer's Caregiver Information Handout

Critique and Revision of the SOMC Email Sequence

The previous email sequence was clear and followed the basic structural requirements. However, the persuasive power, particularly in the Subject Lines and the urgency of Email 3, can be significantly enhanced.

Critique of the Original Sequence

Aspect	Original Content	Critique
Email 1 Subject	Welcome! Ready to start learning?	Friendly, but generic. Could be more focused on the benefit of the platform's content.
Email 2 Subject	Your skill-up is just 10 minutes away.	Good. Directly references the key time benefit for busy
Email 2 Subject	Your skill-up is just 15 minutes away.	Good. Directly references the key time benefit for busy professionals, driving engagement.
Email 3 Subject	Trial ends tomorrow: Don't lose access!	Effective urgency cue, but lacks the specific benefit or curiosity element.
Email 3 Core Message	Trial ends tomorrow: Don't lose access! You're so close to unlocking the medical skills that will set you apart. Don't let your progress stop now...	The message is good but doesn't maximize loss aversion. It needs to explicitly state what they are losing (e.g., specific courses, certificates).

Alzheimer's Caregiver Information Handout

Revised Email Sequence

This revised sequence incorporates benefit-driven subject lines, leverages loss aversion, and adds a compelling, time-bound offer to the final CTA.

Email #	Subject Line	Core Message	Primary CTA
1	How to master a new skill this week.	Welcome to your SOMC trial! You now have seven days of unlimited access to our entire library of expert, practical medical courses. Think about the professional edge you could gain by Friday. Our platform is designed to make expert knowledge quick and accessible for busy professionals like you. Explore the most popular courses and	Unlock Your First Course Now

2

Need 10 minutes to level up?

We noticed you haven't completed your first lesson yet. Our courses are built around your schedule; most lessons take less than 15 minutes to complete. Don't feel you need to find hours!

Just pick one high-value module and see how easily this critical learning fits into your lunch break or commute. Take a quick step forward in your professional development today.

3

Last Chance: Don't lose your access and 20% offer!

Your free trial expires tonight. If you don't upgrade, you will immediately lose access to all specialized courses, professional certificates, and your progress tracking data. Don't let a week of valuable learning go to waste! We want to support your commitment: secure your subscription now and get 20% off your first year. This offer is only available until midnight.

Upgrade for 20% Off (Offer Ends Tonight)

Explanation: The follow-up prompt executed the Iterative Refinement loop by using targeted feedback. The revisions led to:

- 1.More impactful Subject Lines that align with the required Style.
- 2.A stronger closing message in Email 3 that explicitly utilizes urgency cues and a time-bound discount, directly addressing the conversion Objective

e. Framework Effectiveness Reflection (163 words)

The CO-STAR Framework was highly effective for structuring this communication task. It proactively ensured compliance with key Prompting Principles, preventing the output from defaulting to a generic, unpersuasive tone by mandating explicit definitions for Audience, Style, and Tone.

This intentional structure provided the necessary control over the linguistic register and tenor, which are vital for marketing success. It actively mitigated the risk of generating vague answers, a common anti-pattern in prompting.

I would not use a different framework for this specific task. While SMART Prompt is valuable for measuring factual outputs (like quiz generation), it does not prioritize the necessary linguistic controls (e.g., non-technical language, supportive tone) essential for effective customer engagement copy. My key learning is that framework selection must be diagnostic: for human-facing content, frameworks prioritizing register and tenor (like CO-STAR) are mandatory; for data processing or coding, frameworks prioritizing constraints (like RACI) are superior. Matching the framework to the task complexity accelerates process reliability.

Part 3: Critical Reflection (438 words)

a. Skill Progression (135 words)

My understanding of prompt engineering has profoundly evolved from simply treating the LLM as a search engine to viewing it as a sophisticated pattern-matching system that requires precise programming via natural language. My most significant "aha moment" was realizing the power of the Structured Output pattern. Understanding that I could force the model to deliver complex content (like an email sequence) in a clean, predictable table format shifted my perception of the AI from a general text generator to a reliable downstream automation tool.

The most challenging concept was accepting the inherent limitation of Hallucination, particularly its risks in the medical field. I overcame this by adopting Guardrails and Retrieval-Augmented Prompting (RAG) to ensure that output is grounded in provided context, treating every AI response as a "draft" that needs human review.

b. Practical Application (143 words)

A critical real-world scenario in my medical career where prompt engineering could be applied is Automating First-Draft Clinical Notes (e.g., SOAP notes). This would streamline documentation and free up physician time.

Frameworks/Techniques to Use: I would combine the RACI Prompt Canvas with a strong Pattern Stack. The Role would be "Medical Scribe," the Action would be "summarize patient data into a SOAP format," and the Constraints would include the RAG pattern (grounding the output only on provided EHR excerpts) and a Schema Enforcement pattern (ensuring exact SOAP note section headings).

Anticipated Challenges: The primary challenge is Hallucination, which could lead to reporting an incorrect medical history or diagnosis.

Addressing Challenges: I would implement a required Validation Hook in the prompt, forcing the AI to output INCOMPLETE if any required fact cannot be directly cited from the provided patient record.

c. Limitations & Ethics (111 words)

Limitations of AI/LLMs:

- 1.Hallucination:** LLMs are trained to predict the next plausible token, not truth, leading to fabricated facts that are unacceptable in clinical contexts.
- 2.Sensitivity to Phrasing:** LLMs can produce different outputs based on minor prompt changes, making them unreliable unless a consistent framework (like CO-STAR) and versioning are used.
- 3.Ensuring Responsible Use:** I would ensure responsible use by mandating that all outputs include a non-negotiable safety disclaimer (e.g., "This is a draft for physician review only") and strictly enforcing a Refusal Policy if the AI detects a conflict or data gap. This ensures a Human-in-the-Loop (HIL) process is maintained.

d. Goals for the Second Half (49 words)

I aim to master Prompt Chaining to build scalable, multi-step playbooks that go beyond single-turn outputs. Specifically, I want to explore Meta-Prompting to automate the generation, critique, and refinement of specialized prompts for different medical tasks (e.g., generating patient education vs. generating a research summary).